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Ing. Michal Štěpán

Senior GenAI Application Engineer · LLM/RAG/agents · Python/FastAPI · Azure & GCP

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📍 Sviadnov (Ostrava), Czechia · remote / hybrid

🌐 mxstex.github.io/MyWebPage

🗨 Czech native · English B2–C1 (Cambridge FCE)

Profile

Senior GenAI Application Engineer building production AI systems end to end: document extraction and RAG, agent-based assistants, structured LLM validation beside deterministic rules, golden-dataset evaluation, FastAPI backends and cloud delivery. At ABB I work across architecture, quality, backend and operations on Azure; in my own Tojin Games platform I apply the same engineering principles on GCP, including OAuth 2.0, RS256 JWTs and Terraform. Before ABB: freelance backend/data engineering, energy analytics at Siemens and ABB, and eight years in district heating, including technical leadership. Nuclear engineer by training. My next step is broader technical ownership and GenAI architecture.

Production GenAI

LLM/RAG/agents · Azure AI
Foundry · FastAPI

Evaluation first

golden datasets ·
deterministic checks · stress
tests

Agent tooling

Azure AI Agents ·
MCP/FastMCP · Ollama

Cloud architecture

Azure Container Apps · GCP
Cloud Run · Terraform

Core skills

GenAI & LLM applications

Azure OpenAI / AI Foundry · Azure AI Agents
RAG · Azure AI Search · ChromaDB · Document Intelligence
structured outputs · LLM + deterministic rules
golden datasets & evaluation · MCP · FastMCP
LLM agents on Ollama · pgvector memory · Claude Code
GitHub Copilot

Cloud & delivery

Azure Container Apps · Container Registry
Key Vault · managed identity · Log Analytics
Blob & Table Storage · Azure DevOps / Pipelines
GCP Cloud Run · Secret Manager · Terraform (GCP) · Docker
OAuth 2.0 / RS256 JWT

Programming & testing

Python · FastAPI · Pandas · Pytest · SQL / PostgreSQL
Redis · Neo4j · NiceGUI · Vue 3 · Playwright
load & stress tests · NumPy / SciPy

Worked with earlier

LangChain · LangGraph · Dagster · Airflow · DuckDB
BigQuery · RabbitMQ · Kubernetes · GitLab CI
GitHub Actions · Flask · Streamlit · AWS S3 · Lambda
Azure Functions · Grafana · Power BI · Kotlin · WebGL2
ARIMA / SARIMA

Professional experience

Senior GenAI Application Engineer · ABB s.r.o.

Sep 2025 – present

- Build production GenAI applications for industrial engineering workflows: document extraction and RAG, structured LLM validation beside deterministic rules, and agent-based assistants using Azure OpenAI / AI Foundry and Azure AI Agents.
- Own quality and evaluation for AI features: golden datasets, acceptance evaluation, deterministic checks, performance and stress tests, and cloud-vs-local model comparison through Ollama.
- Develop Python/FastAPI backends and application services; use PostgreSQL, Redis and Neo4j where relational, session/cache and graph data are required.
- Deliver through Azure Pipelines across dev, test, stage and prod; Docker images in Container Registry, Azure Container Apps, Key Vault through managed identity and Log Analytics.
- Built a proof-of-concept MCP server (Python, FastMCP) that gives Claude Code and GitHub Copilot controlled access to Azure DevOps work items, with a mock API and tests.

STACK Python · FastAPI · Azure OpenAI / AI Foundry · Azure AI Agents · RAG · ChromaDB · Document Intelligence · PostgreSQL · Redis · Neo4j · Pandas · Pytest · Docker · Azure Container Apps · Key Vault · Azure Pipelines · MCP · Ollama

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Freelance Software / Data Engineer
Sep 2022 – Sep 2025
- DTSE, Landis+Gyr, Vendavo and other clients in energy, telecom and manufacturing · remote, international teams*
- Designed and built end-to-end data workflows in Python, orchestrated with Dagster and Airflow, stored in PostgreSQL, DuckDB and BigQuery.
 - Built backend components for high-volume ingestion and message-based processing with RabbitMQ, and REST services.
 - Migrated services from Kubernetes to cloud-native GCP (Cloud Run, GCS, BigQuery).
 - Maintained CI/CD in GitLab CI and GitHub Actions with unit, integration and end-to-end tests in Pytest.
 - Delivered high-performance processing pipelines for recommender systems (multiprocessing, optimisation); L2 support for production systems.
- STACK** Python · Dagster · Airflow · PostgreSQL · DuckDB · BigQuery · RabbitMQ · GCP · Kubernetes · GitLab CI · GitHub Actions
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Senior Software Developer · Siemens s.r.o.
Jan 2021 – Aug 2022
- Developed an Energy Management System consolidating and analysing data from many sources.
 - Implemented ARIMA and SARIMA models for time-series forecasting.
 - Mentored junior developers.
- STACK** Python · Flask · FastAPI · Docker · PostgreSQL · Grafana · Power BI · RabbitMQ · AWS S3 · AWS Lambda
- 
Software Project Engineer · ABB s.r.o.
Oct 2018 – Jan 2021
- Developed a water and energy consumption evaluation system; designed and implemented predictive-analytics features.
- STACK** Python · Docker · PostgreSQL · Azure Functions · Azure Blob Storage · Grafana · Power BI
- 
IT Specialist · Power Engineer · Technical Deputy · DISTEP a.s.
May 2010 – Oct 2018
- As technical deputy, led an operations team of more than fifteen people in water and heat distribution.
 - Improved resource efficiency through optimised calculations and monitoring; streamlined data collection with PDA barcode readers (MS SQL, VBA).

🍷 Tojin Games · my own studio, in my own time

Independent science, space and simulation games · YouTube youtube.com/@tojin111 · live builds on the portfolio site

- Games:** eight science, space and strategy games built solo, plus an early online prototype of Jednota; five are live in the browser on Google Cloud Run. Hand-written engines for Newtonian gravity, Kepler orbits with WebGL2 rendering, plasma confinement (Biot–Savart) and the Schrödinger equation (split-step Fourier); in Gravity, tests keep the JavaScript engine bit-identical to the Python reference and the Kotlin one within 1e-9.
- Jednota**, a 4X strategy game run by LLM agents (~73,000 lines of Python, about 1,300 tests): strategist, envoy and fleet-commander agents on local LLMs through Ollama, with JSON-schema output and no agent framework; deterministic rules validate every model decision (accepted 93/93 strategic, 80/88 diplomatic, 79/86 tactical in a measured five-empire campaign); memory across games through pgvector RAG.
- Platform:** a FastAPI authentication service completes Patreon OAuth 2.0, maps memberships to internal entitlements and signs short-lived RS256 JWTs. Browser redirects carry only a random one-time code (60-second TTL, stored hashed); game servers exchange it for the JWT. Patreon access tokens are never stored, games hold only the public verification key/JWKS, and Cloud Run reads the private credentials from Secret Manager provisioned with Terraform.
- Video pipeline:** code-driven 4K capture of real game sessions (Playwright, Chrome DevTools Protocol, ffmpeg/NVENC), captions from on-screen events, local TTS, loudness-checked masters and automated validation, plus local ComfyUI models (FLUX, Wan 2.2): more than forty videos published in September 2026.
- How:** AI-assisted development with Claude Code, with the output verified by test suites, golden traces, level verifiers and bots that play the games through.

🎓 Education & certificates

Ing. (MSc), Electrical Engineering and Computer Science · 2012
VŠB – Technical University of Ostrava. Thesis: VBA Utilization for Power Engineering Data Processing and Presentation

Bc., Nuclear Engineering · 2009
Czech Technical University in Prague, Faculty of Nuclear Sciences and Physical Engineering. Thesis: Physical Analysis of Fast Reactor Cores in Up-to-Date Tendencies

Research project · 2006
Institute of Plasma Physics, Czech Academy of Sciences: electrostatic turbulence in the edge plasma of the CASTOR tokamak

Microsoft AZ-305 · AZ-400
Azure infrastructure design and DevOps solutions · training
Cambridge English: First (FCE)
Grade B · CEFR B2 · 2019